

The Impact of Multinational Venture Capital on the Startups' Performance: Insights From Europe

Yang Liu , Yufei Qiu, Fangjing Bu, Wanling Xie, and Lining Han 

Abstract—Venture capital (VC) investment plays a crucial role in the functioning of firms and the overall economy, and its significance has been increasing, with global VC funds reaching new highs in 2023. This study focuses on a sample of European entrepreneurial firms that received at least one VC investment between 1998 and 2015 and examines the impact of multinational VC (MVC) on startups' performance. To ensure a high level of research rigor and the reliability of the results, we address potential selection bias and endogeneity issues by employing the Heckman two-step method and the 2SLS method, with migrant stock serving as the instrumental variable. Our findings reveal that MVC significantly reduces startups' sales and return on assets, particularly in Northern and Central Europe and across both growth and decline periods. Through multiple robustness checks—including alternative model specifications, variable substitutions, and outlier control—we confirm the robustness and reliability of our results. Our findings suggest that startups fare better by seeking VC investment from a single nation.

Managerial Relevance Statement—This study offers practical guidance for engineering managers, startup leaders, and policymakers on how to engage with venture capital (VC) strategically. Contrary to the popular belief that multinational venture capitalists (MVC) always bring superior resources, our findings reveal a surprising insight: startups backed by MVC generally perform worse than those backed by single-country venture capitalists. This unexpected result is primarily driven by governance complexity, cultural frictions, and reduced adaptability, especially during early growth and decline phases. For engineering managers, this insight underscores the importance of evaluating investor composition not only in terms of capital amount but also in terms of strategic alignment. For startup leaders, the findings highlight the need to carefully consider the tradeoffs between accessing international resources and maintaining organizational coherence. They should emphasize strategic fit and governance simplicity when selecting VC partners, especially in the early growth phase, to avoid coordination frictions and performance decline. Managers should

prioritize domestic VC when operating in high-uncertainty environments or during R&D-intensive stages. Policymakers can use these findings to strengthen local capital ecosystems by offering loan guarantees, coinvestment mechanisms, or tax incentives to encourage domestic VC participation, thereby reducing reliance on less-aligned cross-border capital. By integrating regional and life-cycle perspectives, this study provides actionable recommendations to enhance startup resilience and innovation outcomes. It empowers stakeholders to make financing decisions that better align with engineering goals, internal capacities, and long-term performance objectives. This article also contributes to the following SDGs: SDG 8, SDG 9, SDG 12, SDG 16, and SDG 17.

Index Terms—Entrepreneurial firms, multinational venture capital (MVC), startups' performance, syndication.

I. INTRODUCTION

VENTURE capital (VC) is widely recognized as a critical enabling mechanism for young high-tech firms, providing not only financing but also governance, strategic guidance, and monitoring that can shape innovation trajectories and growth outcomes [1], [2], [3]. Global VC fundraising reached approximately \$162.6 billion in 2022, underscoring the increasing reliance of startups on external capital to sustain R&D intensity and organizational scaling [2]. Compared with traditional industrial investment, VC typically involves active equity participation, emphasizes long-term growth, tolerates higher uncertainty, and frequently targets emerging technology sectors—where product development cycles are long and execution risks are substantial. At the macro level, early-stage entrepreneurship contributes to job creation and innovation-driven economic expansion [4], [5], making the performance consequences of VC involvement a central question for both scholars and practitioners.

Despite extensive research on VC, we still know relatively little about how international syndication affects the day-to-day managerial and engineering realities of startups. In particular, the interaction dynamics among investors from multiple countries—whether they reinforce each other's value-added services or generate coordination and control frictions—remain underexplored [6]. This gap is consequential for engineering management because multinational venture capital (MVC) can reshape project governance and technology execution: cross-border syndication often requires firms to reconcile heterogeneous expectations regarding reporting standards, milestone definition, decision rights, and post-investment intervention. Macroeconomic factors in the investor and portfolio firm countries, alongside deal and investor characteristics, may further alter the costs

Received 19 March 2025; revised 17 July 2025, 7 October 2025, 15 November 2025, and 29 January 2026; accepted 23 February 2026. Date of publication 26 February 2026; date of current version 27 March 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 72574174 and Grant 72473109, in part by the Youth Fund for Humanities and Social Science Foundation of the Ministry of Education of China under Grant 22YJZH114 and Grant 19YJC790033, and in part by the Scientific research project of the National Language Commission of China under Grant YB145-74. Review of this article was arranged by Department Editor B. Shen. (Corresponding author: Lining Han.)

Yang Liu is with the School of Information Management, Wuhan University, Wuhan 430072, China.

Yufei Qiu is with the School of Accountancy, Zhongnan University of Economics and Law, Wuhan 430073, China.

Fangjing Bu, Wanling Xie, and Lining Han are with Economics and Management School, Wuhan University, Wuhan 430072, China (e-mail: hanlining@gmail.com).

Digital Object Identifier 10.1109/TEM.2026.3668200

and benefits of cross-border collaboration [7]. Compared with single-country VC, MVC is more likely to involve higher communication costs due to information asymmetry and cultural distance, slower decision cycles due to dispersed principals, and greater governance complexity as firms navigate multiple institutional environments and compliance requirements [8]. By contrast, domestic VC—particularly domestic government-backed funds (GVCs)—typically operates within familiar regulatory regimes, aligns with local innovation and development goals, and tends to follow more standardized governance practices that can reduce managerial friction and coordination cost [9].

Conventional wisdom in venture finance often emphasizes the benefits of cross-border capital—access to global networks, complementary expertise, and international market linkages—implicitly suggesting that more international investor bases should improve performance. From an engineering management perspective, however, these potential benefits are not automatic: the value of external capital depends on whether the firm can translate investor resources into effective resource orchestration, fast iteration, and coherent technology-roadmap execution. MVC may introduce hidden “execution taxes” such as misaligned technological priorities, conflicting milestone logic, duplicated monitoring demands, and governance deadlocks that slow product development and reduce operational efficiency. If MVC systematically undermines performance, this would revise a key assumption underlying both VC practice and engineering-management theory: internationalization of investment does not necessarily enhance innovation execution; instead, its net effect is contingent on governance complexity and the firm’s coordination capacity.

To examine this issue, we empirically estimate how MVC affects startup outcomes using the VICO 2.0 database, focusing on sales and return on assets (ROA). Europe provides an ideal setting because its VC landscape is heterogeneous, characterized by institutional diversity and uneven capital distribution [15], [16], which allows us to observe how governance frictions vary across regional contexts. Methodologically, we implement a multistep identification strategy to address selection and endogeneity concerns. We first apply propensity score matching to mitigate observable self-selection into MVC, then employ Heckman’s two-step model to correct for potential selection on unobservables, and finally estimate 2SLS models using migrant stock as an instrument for MVC exposure. We further validate robustness through alternative variable definitions and the coarsened exact matching (CEM) method.

Our findings indicate that MVC is associated with significantly lower startup performance in both sales and ROA relative to single-country VC, and this pattern remains stable across alternative identification approaches. Importantly, the negative association is heterogeneous: it varies across European regions and across firm development stages. The performance penalty is stronger in Northern and Central Europe and is more pronounced for firms in growth and decline stages—consistent with the view that institutional and governance complexity amplifies coordination frictions when firms have limited organizational slack. These results extend engineering management research by shifting attention from the average benefits of external capital to the governance–execution trade-off embedded in cross-border

syndication: when coordination costs dominate, MVC can hinder innovation execution and operational performance rather than strengthen it.

The remainder of the article is organized as follows. Section II reviews related literature and develops hypotheses. Section III presents the data, methodology, and empirical models. Section IV reports baseline findings and robustness checks. Sections V and VI discuss implications, limitations, and conclusions.

II. THEORY AND HYPOTHESES

A. Multinational Venture Capital and Startup Performance

VC is widely acknowledged for enhancing startup performance by alleviating financial constraints and fostering innovation through R&D investment, patent generation, and productivity improvements [21]. However, some studies suggest that these positive effects vary by region and investor type, with limited or insignificant outcomes in certain national contexts [23]. MVC, a growing form of cross-border VC, introduces both opportunities and challenges. On one hand, foreign VC investors can legitimize unfamiliar startups [12], help international expansion [29], and provide access to strategic global resources, driven not only by financial returns but also by strategic and noneconomic motives [13], with these effects moderated by the venture’s learning proficiency [14]. On the other hand, they may generate coordination difficulties, cultural frictions, and information asymmetries [15]. Unlike single-country VC [16], which benefits from local familiarity and streamlined operations, MVC must navigate diverse legal systems, institutional environments, and regulatory frameworks [17].

Compared to single-nation venture capital, MVC exhibits fundamental structural and strategic differences that shape its impact on startup performance [18]. Single-country VC investors—whether private or government-affiliated—typically operate within unified institutional settings, share cultural familiarity with the investee firms, and follow well-aligned governance practices. These advantages reduce coordination costs, improve trust and communication, and enable quicker, more responsive decision-making [19]. In contrast, MVC involves actors from multiple countries who must coordinate across different legal systems, time zones, languages, and strategic priorities, introducing greater uncertainty, monitoring challenges, and potential strategic misalignments [20]. These differences form a core basis for examining the heterogeneous effects of MVC on startups.

While some research suggests that MVC can promote long-term growth [21], others highlight its negative effects on exit prospects, innovation, and resource allocation efficiency [22]. These conflicting findings call for further empirical investigation into the performance outcomes of startups supported by MVC. Three theoretical perspectives explain the potential drawbacks of MVC [23]. Transaction cost economics (TCE) suggests that the geographic and legal complexity of MVC increases governance costs, diverting attention from core innovation. Resource dependence theory (RDT) posits that the globalized resource structures of MVC may not align with local startup needs [24], weakening firms’ ability to leverage contextual advantages.

Dynamic capabilities theory (DCT) further indicates that MVC can constrain a startup's capacity to sense and seize opportunities due to misaligned incentives, reduced autonomy, and investor path dependence [19]. Given these constraints, startups backed by multinational VC may face performance disadvantages compared to those supported by domestic VC. Therefore, we propose:

Hypothesis 1: Startups receiving investments from VC with multiple nationalities experience lower performance compared to startups backed by single-nationality VC.

B. Heterogeneous Effects of MVC on Startup Performance Across Regions

Significant regional disparities in economic and entrepreneurial dynamics exist across Europe, particularly between Northern and Southern regions [25]. Prior studies have highlighted differences in per capita income, innovation capacity, and industrial development [26]. However, limited research has examined how such regional variation influences the impact of venture capital, particularly in the context of MVC. Drawing on institutional theory and RDT, we argue that the effects of MVC are not uniform across regions. Institutional theory emphasizes the role of formal and informal institutional environments in shaping firm behavior and performance [27]. In regions with weaker institutions or more volatile regulatory environments, such as parts of Southern Europe, the liabilities of foreignness and coordination complexity associated with MVC may be exacerbated. Meanwhile, RDT suggests that startups rely heavily on external resources and must adapt to their availability and structure. MVC, with its globally structured resources, may be misaligned with the localized needs and institutional contexts of startups, thereby affecting their ability to effectively leverage these investments.

Addressing this gap, our study makes two contributions. First, we move beyond prior work focused on specific VC types such as institutional or governmental VC [28], and analyze a broader spectrum of venture capital configurations. Second, we systematically investigate how the performance impact of MVC varies across Northern, Central, and Southern Europe, capturing the interaction between venture capital structure [29] and regional institutional environments [30]. By integrating investment type with geographic and institutional context, this research uncovers important variations in how MVC affects startups across Europe. These findings offer theoretical and practical insights for policymakers, investors, and startup managers. Accordingly, we propose:

Hypothesis 2: The impact of MVC investments on startups' performance varies across different geographical regions.

C. Heterogeneous Effects of MVC on Startup Performance Across Developmental Stages

The performance impact of external investment varies significantly across a startup's development stages. Existing literature highlights systematic differences in financing needs and strategic priorities at different phases of the startup lifecycle—typically

categorized as growth, maturity, and decline [31]. During the growth stage, startups prioritize rapid market expansion and product iteration [32], relying heavily on external funding to sustain R&D and operations. However, limited internal integration may reduce efficiency and responsiveness. In contrast, mature firms possess stable business models and cash flows [33], focusing on operational standardization and capital optimization [34]. Declining firms, often facing market saturation or technological disruption, require strategic transformation under conditions of increased rigidity and path dependence [35].

Theoretical frameworks such as lifecycle theory [36] and dynamic capabilities theory [64] emphasize the need for stage-specific resource configurations. While MVC offers capital and global networks, its complex governance and coordination demands may hinder agility in early-stage firms [31]. Regulatory compliance and extended approval cycles can delay critical decisions, particularly in hardware startups requiring fast iterations. Cultural and institutional mismatches may further impede team cohesion and knowledge sharing [32].

In mature firms, however, established internal systems can better absorb MVC resources, facilitating global expansion and technological transfer. Prior research also suggests that such firms can reduce governance challenges through localized adaptation strategies [34]. Conversely, in the decline phase, the restrictive terms often attached to MVC—such as liquidation preferences—can intensify agency conflicts and reduce strategic flexibility [36]. In summary, due to varying market demands, resource dependencies, and governance capacities, the effect of MVC on startup performance is likely to differ across different stages of development. Thus, we propose:

Hypothesis 3: The impact of MVC investments on startups' performance varies depending on the stage of the company's development.

III. DATA AND MATERIALS

A. Data Collection

We obtained financial data on high-tech startups from the VICO 2.0 database.¹ This database includes companies that have received at least one venture capital or angel investment from 1998 to 2015 and are located in seven European countries (Belgium, Finland, France, Germany, Italy, Spain, and the United Kingdom), as well as Israel. The dataset is unique in terms of its total number of companies (17 863), total number of investors (7834), country coverage, and the range of information collected. The data in our study can be categorized into three primary levels: company-level data, investor-level data, and investment-level data. Thorough information is compiled for each company, investor, and investment, encompassing company financial details, investor categorization, and specific information related to each investment round, such as the number of rounds completed.

To expand our analytical scope, we integrated additional data sources. Specifically, we incorporated data on international migrant stock from The Global Migration Data Portal. This

¹[Online]. Available: <http://datasets.ris.is.eu/metadata/vico>

TABLE I
INFORMATION OF THE INDUSTRIES OF THE STARTUPS

Company Industry	Freq.	Percent
Chemicals	105	0.59
Construction & real estate	294	1.65
Consultancy	845	4.73
Culture & entertainment	410	2.3
Energy & environment	138	0.77
Financial services	605	3.39
Food	221	1.24
High-tech manufacturing	830	4.65
Life sciences	1193	6.68
R&D and engineering	1233	6.9
Software & internet	4009	22.44
Support services	870	4.87
Telecommunication services	333	1.86
Wholesale & retail trade	1052	5.89
Other	5725	5.66
Total	17 863	100

Note: The table shows the information of the company industry. In this table, there are 15 types in total. Freq means the number of companies that belong to this group. Percent represents the corresponding value as a percentage of the total value.

supplementary data gauges the extent of countries' openness and international engagement. The dataset illustrates the proportion of international migrant stock to the total population in each country, computed over five-year intervals from 1990 to 2020. To align with the temporal scope of the investment data within the VICO database, we focused on the period from 1998 to 2015.

Due to the nonavailability of migrant stock data for every individual year, we employed a linear interpolation technique spanning five-year periods. Specifically, we calculated an arithmetic progression within each quinquennial interval, and any absent data points were subsequently substituted with the corresponding values derived from the interpolation. This procedure generated yearly migrant percentage data between 1998 and 2015 for the seven European countries under investigation, ensuring consistency with the firm-level investment dataset.

B. Data Description

Prior research has demonstrated that industry conditions significantly shape entrepreneurial outcomes, as performance declines more steeply for experienced founders in high-tech sectors, credit risk is more detrimental in high-clockspeed industries, and even perceived competitive advantage can increase failure rates in low-tech contexts, highlighting the need to account for industry-specific effects in venture performance studies [37]. Table I summarizes the industry distribution of sampled startups across 15 categories, with software and internet companies comprising the largest share (22.44%), followed by R&D and engineering. The original database categorizes firms based on international standard classification systems (NACE Rev.2/NAICS). For analytical purposes, we re-grouped these categories into 15 broader sectors, highlighting industries most relevant to venture capital research (e.g., software & internet, life sciences, high-tech manufacturing, and R&D & engineering), while aggregating less represented categories into "Other." To make the distribution more relevant for venture capital

research, sectors with substantial representation in the startup ecosystem—such as software & internet, life sciences, high-tech manufacturing, and R&D & engineering—are reported separately. At the same time, smaller categories are grouped into "Other." In the regression analyses, industry heterogeneity is further controlled by including industry fixed effects, ensuring that differences in sectoral composition do not drive results.

Table II details the geographic distribution of firms. After excluding Israel and countries with fewer than five firms (e.g., Luxembourg, Netherlands, Sweden, Switzerland), we focus on startups from seven European countries, grouped into Northern, Central, and Southern Europe for regional analysis.

C. Variables

1) *Dependent Variables*: Firm performance is measured using two indicators. First, $\ln(\text{Sales})$, the logarithm of net sales, captures firm size and market expansion. This transformation reduces heteroskedasticity and allows comparability across firms of different scales. Second, ROA reflects financial efficiency and profitability, measuring how effectively firms generate earnings from their assets. The combination of these two variables ensures that both scale-oriented and efficiency-oriented aspects of firm performance are captured, consistent with prior studies on venture capital impacts [2].

2) *Independent Variable*: The main independent variable is Multiple_it , a binary indicator equal to 1 if firm i received venture capital from more than one country by year t , and 0 otherwise. This variable captures the potential advantages of cross-border venture capital, including knowledge spillovers, diversified resources, and international market access. Following prior literature, we expect that firms backed by multicountry venture capital may exhibit distinct performance outcomes compared to those with single-country VC.

3) *Instrumental Variable*: To address potential endogeneity concerns regarding the relationship between multicountry venture capital and firm performance, we employ Migrant,

TABLE II
INFORMATION OF THE STARTUPS' NATIONS

Region	Company Nation	Freq.
Northern Europe	Belgium	557
	Finland	1044
	United Kingdom	5861
Central Europe	France	3868
	Germany	3094
Southern Europe	Italy	738
	Spain	1454

Note: This table shows detailed information about specific classifications in terms of geographical positions. Northern Europe includes three countries, Belgium, Finland, United Kingdom, respectively. Regarding Central Europe, it covers France and Germany. Italy and Spain are classified in Southern Europe.

TABLE III
VARIABLE DESCRIPTIONS

Variable	Description
ln(Sales)	The logarithm of net sales.
ROA	The return on assets.
Multiple	1 if company i has been backed by VC from different nations by the year t .
Age	The year since the company was founded.
IF to TA	Intangible on total assets.
Cashflow	Profit for period + Depreciation.
Loans	Short-term financial debts to credit institutions.
Shareholder funds	Total Equity (Capital + Other shareholders funds).
Migrant	International migrant stock as a percentage of the total population.

measured as the share of international migrant stock in the total population, as an instrumental variable.

The rationale for this choice is twofold. First, higher levels of international migration in a country are positively associated with the likelihood of attracting foreign venture capital, since migrant networks often facilitate cross-border financial flows, entrepreneurial activities, and information exchange. This provides the required correlation between the instrument and the endogenous variable, *Multiple_it*.

Second, the share of international migrants in the general population is unlikely to directly affect an individual firm's sales or profitability, except through its influence on access to multi-country venture capital. This satisfies the exclusion restriction, as migrant stock reflects macro-level demographic patterns rather than firm-level operational or financial outcomes.

Data on migrant stock are obtained from [insert source, e.g., the World Bank's World Development Indicators or United Nations International Migrant Stock database], ensuring consistency and reliability.

4) *Control Variables*: To isolate the impact of cross-border venture capital, we introduce a set of firm-level controls reflecting financial structure and operational capacity. Age: the number of years since firm founding, controlling for firm maturity and lifecycle stage. Cashflow: calculated as profit for the period plus depreciation, reflecting internal financing ability. IF to TA (Intangibles-to-Total Assets ratio): the share of intangible assets over total assets, capturing the firm's reliance on intangible capital. Loans: short-term financial debts to credit institutions, indicating leverage and external financing dependence. Shareholder funds: total equity, reflecting the

solvency of ownership structure and capital base. These controls are selected because they directly affect firms' growth potential and profitability. For example, age and shareholder funds proxy for organizational maturity and operational resilience, while cashflow, loans, and intangible intensity reflect financing constraints, leverage, and asset composition. Their inclusion ensures robustness of estimation and comparability with related studies.

The study adopts two dependent variables: ln(Sales) (logarithm of net sales) and ROA, reflecting firm size and financial performance, respectively. Table III provides definitions of all variables, including the key independent variable, *Multiple*, which equals 1 if the firm is backed by VC from multiple countries.

D. Methodology

1) *Data Processing*: We measure firm performance using two dependent variables: the logarithm of net sales (ln(Sales)) and ROA [69, 73]. The main independent variable, *Multiple_it*, is a binary indicator equal to 1 if firm i received venture capital from more than one country by year t , and 0 otherwise. Control variables include firm age (measured as years since founding), cash flow (profit plus depreciation), intangible assets ratio (IF to TA, intangible assets divided by total assets), loans (short-term financial debts to credit institutions), and shareholder funds (total equity including capital and other shareholder contributions). Together, these variables capture the financial structure and operational capacity of firms (see Table III).

To ensure accuracy and reproducibility, several data treatment procedures were applied. Specifically, all continuous financial

variables were minorized at the 1st and 99th percentiles to reduce the influence of outliers, and financial data were standardized into euros using contemporaneous annual exchange rates from the European Central Bank. In addition, logarithmic transformation was applied to skewed variables such as Sales to mitigate heteroskedasticity, while industry and country codes were harmonized according to the NACE Rev. 2 classification to enhance comparability across firms. Finally, a firm-year panel dataset was constructed by aligning observations with the investment year, ensuring consistency between dependent, independent, and control variables.

The construction of the Multiple_{it} variable was carried out using Python. Firm-level investment records were first merged with investor-country identifiers. For each firm-year, we applied a group by operation in Python (pandas) to count the number of distinct investor countries. If the count was greater than one, Multiple_{it} was coded as 1 (MVC); otherwise, it was coded as 0 (single-country VC). This automated step ensures accuracy and replicability in distinguishing between multicountry and single-country venture capital backing.

These steps collectively reinforce the robustness and validity of the empirical design.

2) *Rationale for Methodological Choices*: MVC is not randomly assigned: startups receiving multinational VC may systematically differ from those funded by single-country VCs, creating concerns of self-selection and endogeneity. We therefore adopt a triangulated strategy to improve interpretability. First, propensity score matching (PSM) enhances comparability by balancing observable pre-investment characteristics across groups. Second, Heckman's two-step model addresses potential selection on unobservable that may jointly affect the likelihood of receiving MVC and subsequent performance. Third, we estimate 2SLS models using migrant stock as an instrument for MVC exposure, leveraging the idea that migrant networks facilitate cross-border investor-entrepreneur matching and information flow (relevance), while not directly affecting firm performance conditional on controls and time/region fixed effects (exclusion).

In addition, we conduct robustness checks to reduce sensitivity to modeling and measurement choices. We apply CEM as an alternative matching approach that relies on transparent coarsening and exact matching within strata, providing a stricter and more interpretable balance procedure than propensity-score-based matching. We also re-estimate key specifications using alternative operationalizations of the main explanatory variables and outcome measures to ensure that a particular definition or measurement of MVC and performance does not drive the results. The consistency of findings across these methods strengthens the credibility and accessibility of our empirical conclusions.

3) *Propensity Score Matching (PSM)*: To ensure the credibility of our causal inferences, we adopt a rigorous multistage econometric strategy designed to isolate the true impact of MVC from potential confounding factors. Specifically, the relationship between MVC and startup performance is often obscured by two primary challenges: self-selection bias and endogeneity. The former arises because high-quality startups may inherently attract multinational investors (the "picking winners" effect), while the latter may stem from omitted regional variables or

reverse causality. To mitigate these concerns, we first utilize Propensity Score Matching (PSM) and CEM to ensure that MVC-backed firms are compared with domestic-backed firms sharing similar pre-investment characteristics. Furthermore, we implement Heckman's two-step procedure to correct for potential non-random selection in the investment process. Most importantly, we address endogeneity via Two-Stage Least Squares (2SLS) estimation. We employ migrant stock as an instrumental variable, predicated on the logic that historical migration networks facilitate cross-border capital flows by reducing information asymmetry, yet they do not directly influence the contemporary operational efficiency or ROA of an individual startup. This multipronged approach strengthens the internal validity of our findings and provides a more transparent roadmap for readers to evaluate our empirical rigor.

4) *Propensity Score Matching (PSM)*: To reduce self-selection bias in MVC assignment, we implement propensity score matching. For each firm receiving MVC, we identify a comparable firm backed by domestic VC with a similar probability of receiving MVC. The probability is estimated using a probit model based on firm-level characteristics (industry, year, financials). Matching is conducted using nearest neighbor matching with Mahalanobis distance in a 1:1 ratio, with replacement. Following study [38], the bias from replacement remains minimal.

In the second stage, an OLS regression incorporates the inverse Mills ratio (IMR) to correct for selection bias.

5) *Heckman Two-Step Model*: Given the nonrandom nature of VC investment, we employ the Heckman correction model [39] to address selection bias and potential endogeneity. The first stage estimates the likelihood of receiving MVC using a probit model:

Selection model:

$$\begin{aligned} \text{Multiple}_{it} = & \gamma_1 + \gamma_2 \text{Migrant}_{it} + \gamma_3 T_t + \gamma_4 \text{Industry}_i \\ & + \gamma_5 \text{IF to TA}_{it} + \gamma_6 \text{Age}_{it} + \gamma_7 \text{Cashflow}_{it} \\ & + \gamma_8 \text{Loans}_{it} + \gamma_9 \text{Shareholders funds}_{it} + \varepsilon_{it}. \end{aligned} \quad (1)$$

In the second stage, an OLS regression incorporates the IMR to correct for selection bias:

Outcome model:

$$\begin{aligned} \text{Performance}_{it} = & \beta_0 + \beta_1 \text{Multiple}_{it} + \beta_2 \text{IMR}_{it} + \beta_3 T_t \\ & + \beta_4 \text{Industry}_i + \beta_5 \text{IF to TA}_{it} + \beta_6 \text{Age}_{it} \\ & + \beta_7 \text{Cashflow}_{it} + \beta_8 \text{Loans}_{it} \\ & + \beta_9 \text{Shareholders funds}_{it} + \varepsilon_{it} \end{aligned} \quad (2)$$

where $\text{Performance}_{it} \in \{\ln(\text{Sales}_{it}), \text{ROA}_{it}\}$.

The Migrant variable, reflecting the international openness of a country, is used only in the selection model, as it is country-level and unlikely to directly affect firm-level performance.

6) *Two-Stage Least Squares (2SLS) Regression*: To further address endogeneity, we conduct a 2SLS estimation using Multiple_{it} as an instrumental variable for Multiple_{it} . Migrant stock is assumed to affect the probability of receiving MVC, but not directly influence firm performance conditional on controls.

The structural model estimated is as follows:

$$\begin{aligned} \text{Performance}_{it} = & \beta_0 + \beta_1 \text{Multiple}_{it} + \beta_2 T_t + \beta_3 \text{Industry}_i \\ & + \beta_4 \text{IF to TA}_{it} + \beta_5 \text{Age}_{it} + \beta_6 \text{Cashflow}_{it} \\ & + \beta_7 \text{Loans}_{it} + \beta_8 \text{Shareholders funds}_{it} \\ & + \varepsilon_{it}. \end{aligned} \quad (3)$$

This approach provides a robustness check on the Heckman model and strengthens causal interpretation by using an external instrument.

IV. RESULTS

A. Regional Heterogeneity

Table IV shows the regression results utilizing the OLS, Heckman two-step method, and 2SLS method. The key independent variable is the dummy variable Multiple_{it} , which reflects whether a startup has received VC investments from multiple nations. Meanwhile, the accounting variables such as $\text{Shareholders funds}_{it}$, Cashflow_{it} , Loans_{it} , Age_{it} , as well as year fixed effects and the industry fixed effects are added as the control variables. Columns (1) to (3) take $\ln(\text{Sales})$ as the proxy of the company's performance, while columns (4) to (6) take ROA, the return on assets, as the proxy variable. As the estimates shown in columns (1) and (3), with the OLS methods employed, no matter choosing the logarithm of the sales or ROA as the dependent variable, the MVC has a significantly negative effect on the performance of startups. The same result occurs when utilizing the Heckman two-step method, according to columns (2) and (4). Similarly, the 2SLS regression results in columns (3) and (6) show that the MVC has a significantly negative effect on performance, whether measured by sales or return on assets.

Note that in columns (1), (2), and (3) of Table IV, where the dependent variable is defined as $\text{Performance}_{it} = \ln(\text{Sales}_{it})$, the regression coefficients of Multiple_{it} are successively -0.204 , -0.216 , and -1.437 . These negative coefficients indicate that, compared with the baseline of single-country VC ($\text{Multiple}_{it} = 0$), startups backed by multicountry VC ($\text{Multiple}_{it} = 1$) experience lower sales and profitability, suggesting a relative disadvantage of multicountry VC. Similar trends are observed when comparing columns (4)–(6). This pattern can be attributed to the possible presence of endogeneity and selection bias. As investors base their investment decisions on their expectations of startups, companies possessing certain attributes that suggest promising prospects are more likely to attract VC investments from different countries. Furthermore, the performance of companies can also impact their likelihood of being included in the VICO dataset. Well-performing companies might enjoy greater visibility and willingness to disclose their operational status. Consequently, the coexistence of selection bias and endogeneity could lead to an upward bias in the regression estimation.

The findings of this study emphasize the potential impact of disparate management approaches rooted in diverse cultural contexts [40]. Such differences may lead to conflicting dynamics that could hinder operational efficiency within management processes. This contrast becomes more apparent when juxtaposed with the operational harmony characteristic of systems from a

single cultural background. This observation holds noteworthy implications, suggesting that during the pivotal startup phase, the strategic decision to align with venture capital investments originating from a culturally homogeneous context might yield more significant advantages for the overall advancement of the company.

B. Further Considerations

1) *Heterogeneity of the Regions:* To explore regional variation in the impact of MVC, we divide the sample into Northern Europe (Finland, Belgium, U.K.), Central Europe (France, Germany), and Southern Europe (Italy, Spain), and apply the Heckman two-step model separately to each group. Tables V and VI show the results of OLS and Heckman two-step, respectively. While the estimates mainly coincide with the previous results shown in Section V, there exists regional heterogeneity. The results show that the MVC has a significantly negative impact on the performance of startups in both Northern and Central Europe, whereas the effect is insignificant for startups in Southern Europe.

The negative impacts in Northern and Central Europe can be explained by the differences in institutional environment, corporate governance, and investor-founder dynamics. Cross-border venture capital investments usually involve investors from different legal, cultural, and economic backgrounds, which may lead to inconsistencies in strategic objectives and management styles. In Northern and Central Europe, the corporate governance structures are more stringent, and the regulatory frameworks implement stricter supervision. The participation of foreign venture capital investors may exacerbate the conflicts between founders and investors, reducing operational efficiency [41]. Wu et al. [42] found in their research that the banking systems in Northern and Central European countries tend to rely on traditional credit models. After the entry of cross-border capital, due to the conflicts with local financial regulations, it has instead intensified the financing constraints of enterprises.

In contrast, the business and institutional environments in Southern European countries (such as Italy and Spain) are more informal and flexible, which may be conducive to the smoother integration of cross-border venture capital. The case study [43] shows that the institutional settings of different countries will affect the local embeddedness of entrepreneurial families in different ways. The investment practices based on trust and the greater tolerance of Southern European enterprises toward the influence of foreign management can reduce the conflicts related to the participation of MVC, thus having a less obvious negative impact on corporate performance. In sum, the performance impact of MVC varies across regions, reflecting differing levels of institutional rigidity, cultural adaptability, and regulatory alignment. These findings underscore the need to account for geographical context when evaluating the effects of cross-border VC.

2) *Heterogeneity in the Stages of Firm Development:* Furthermore, considering that the impact of MVC from different countries on the performance of startups may vary depending on the stage of the firm's development, we conduct a comprehensive analysis. There are diverse approaches to classifying

the corporate life cycle, among which the comprehensive score discrimination method is most applied [44]. Research indicates that ventures face complex tradeoffs in resource acquisition decisions, which have implications for multiple milestones across a venture's lifecycle. In this study [45], we adopt this widely recognized method and accurately measure the development stage of a company by examining four key variables: firm age, revenue growth rate, retention earnings rate, and capital expenditure rate. To delve deeper, we first divide the startups into three groups according to their development stages - growth stage, mature stage, and decline stage. Then, we separately apply the grouped regression using both OLS and Heckman's two-step method to these three groups of samples. This is to explore whether the combination of VC has different effects on startups at various development stages [42]. To further address potential sectoral heterogeneity, industry fixed effects are included in all regressions. Finally, we conduct the regression process on the entire dataset, which encompasses all startups regardless of their development stage, to further validate the overall relationship and potential differences.

Tables VII and VIII present the results of grouped regression tests using both OLS and Heckman's two-step methods across different stages of enterprise development. The regression results reveal stage-specific heterogeneity in the impact of the key independent variable $Multiple_{it}$ on firm performance. In models using the logarithm of sales ($\ln(\text{Sales})$) as the performance measure, the coefficients for $Multiple$ were significantly negative for firms in both the growth and decline stages (OLS coefficients of -0.206 and -0.133 , respectively; Heckman coefficients of -0.199 and -0.129). This suggests that multicountry VC investments have a pronounced inhibitory effect on firm performance during these stages. However, for firms in the mature stage, the effect appeared slightly significant in the OLS model (with a coefficient of -0.100) but was not significant in the ROA model (OLS coefficient of -0.020 ; Heckman coefficient of -0.022). This heterogeneity may stem from differences in internal governance and organizational maturity across development stages: firms in the growth stage, due to limited management experience, are more susceptible to management conflicts arising from diverse investments, while firms in the decline stage may be more sensitive due to operational challenges. In contrast, mature firms, with their well-established internal mechanisms, are better equipped to absorb and mitigate the impact of external investments. Startups are often characterized by limited managerial and marketing experience, which makes them less attractive to most venture capital firms that typically prefer to invest in more mature enterprises [46]. In sum, the analysis confirms that MVC's performance impact is stage-dependent, being most detrimental during growth and decline phases. These findings underscore the importance of aligning investment strategies with the recipient firm's lifecycle stage, as firms' absorptive capacities and governance readiness evolve over time [34].

C. Robustness Checks

We conducted many robustness tests on the empirical results. In the data processing stage, the sample selection bias was dealt with through propensity score matching (PSM). To address the

endogeneity problem, the Heckman two-step method was used in the main regression, and the 2SLS method of introducing the migrant instrumental Variable was employed to deal with the endogeneity problem caused by omitted variable bias or reverse causality. Furthermore, we also conducted the substitution of independent variables and the robustness test of the CEM method. These results were all consistent with the main regression results, confirming the hypothesis.

1) *Alternative Independent Variable*: In the previous sections, we utilized the binary variable $Multiple_{it}$, which takes on values of 0 or 1, to represent MVC investments. However, this binary variable may not fully capture more specific information, such as the level of diversification within MVC investments [39]. To address this issue, we conducted new regression analyses, replacing $Multiple_{it}$ with the variable $Count_{it}$, which represents the natural logarithm of the count of unique countries from which company i received investments in year t from its historical data.

Incorporating $Count_{it}$ as the independent variable and replacing $Multiple_{it}$, we re-estimated the regression models. The OLS regression results are presented in columns (1) and (3) of Table IX, and the Heckman regression results are presented in columns (2) and (4). The significantly negative coefficients observed for $Count_{it}$ suggest that the adverse impact of MVC investments becomes more pronounced as the diversity of countries contributing to the investments increases. This finding reinforces the conclusions drawn from our earlier estimations.

2) *Coarsened Exact Matching (CEM)*: The previous analysis employed PSM, an "equal percent bias reducing" matching method that does not guarantee imbalance reduction in every dataset. To address this, we use CEM, a "Monotonic Imbalance Bounding" method that operates without assumptions about the data-generating process [41]. The CEM algorithm proceeds by creating a coarsened copy of covariates, binning them into strata, and retaining only those strata containing both treated and control units [42]. Using the binary variable $Multiple_{it}$ as treatment, startup size $Sales_{i,t-1}$ and age as pre-treatment covariates, we obtained 194 matched strata (out of 268), covering 64 227 of 64 269 treated and 92 342 of 93 265 control samples. The CEM process also generated the weights applied in subsequent 2SLS regressions, with results (see Table X) consistent with prior analyses.

V. DISCUSSIONS AND IMPLICATIONS

A. Implications for Research

This study contributes to the literature on venture management and engineering management in three ways.

- 1) First, we integrate TCE, RDT, and DCT to clarify why MVC can hinder startup performance. TCE emphasizes rising governance and coordination costs when multiple foreign investors must align monitoring and decision rights. RDT highlights that dependence on diverse external investors can constrain technological autonomy and strategic discretion. DCT suggests that navigating multiple institutional contexts can strain a young firm's adaptive capacity, reducing agility when rapid pivots are required [24].

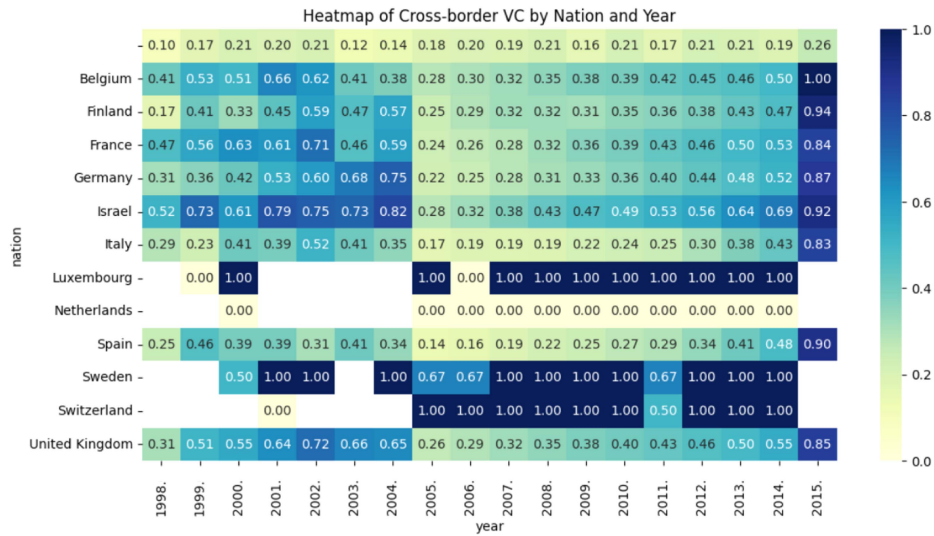


Fig. 1. Heatmap of cross-border VC by nation and year.

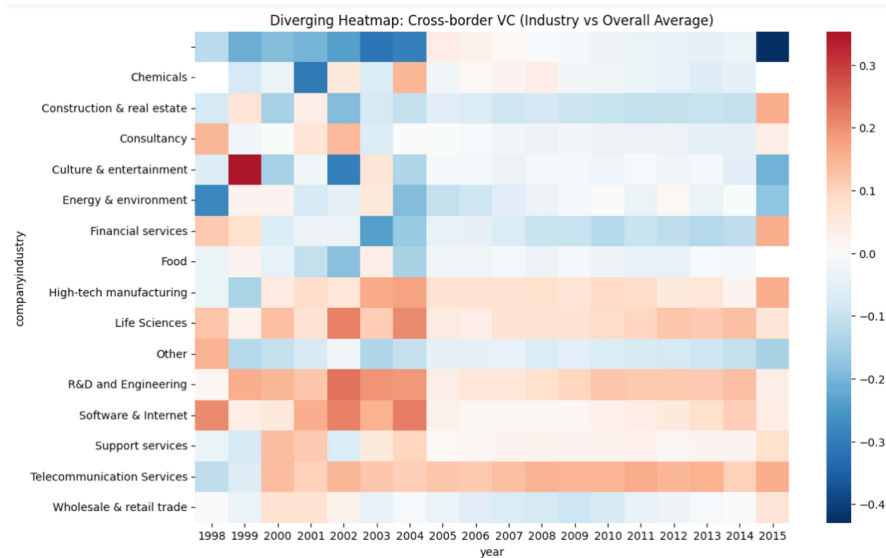


Fig. 2. Diverging heatmap: cross-border VC (industry versus overall average).

- 2) Second, consistent with institutional theory [47], we show that MVC's performance impact varies across European regions. Institutional distance can translate into higher cross-border transaction costs, and these frictions may be particularly salient in environments with strong formal institutions, where compliance demands and governance expectations are more stringent—raising the cost of alignment with foreign investors [48].
- 3) Third, extending life cycle theory [49], we demonstrate that firm development stage moderates the MVC–performance relationship. Growth-stage and decline-stage firms appear more vulnerable to MVC-related disruptions (e.g., slower decisions, governance frictions), whereas mature firms with established routines and internal systems can better absorb external capital without comparable performance penalties.

B. Implications for Practice

The regression results obtained from this study have valuable implications for start-ups, investors, and European countries.

First, as the VC industry matures, startups face more significant investment opportunities, making selecting the most suitable VC investments increasingly challenging [2]. Given our finding that MVC investments harm startups, startups should prioritize seeking venture capital from investors within the same nation. This choice will likely yield better outcomes for startups in terms of performance and growth.

For startup leaders, this finding underscores the importance of balancing access to international resources with the need for strategic and organizational coherence. They should carefully assess investor compatibility in terms of governance structure, cultural fit, and communication efficiency to avoid coordination

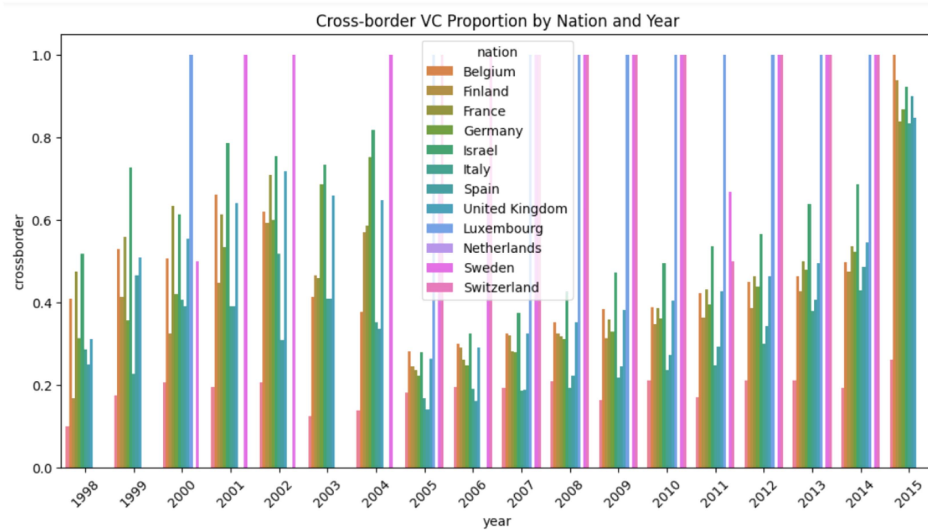


Fig. 3. Cross-border VC proportion by nation and year.

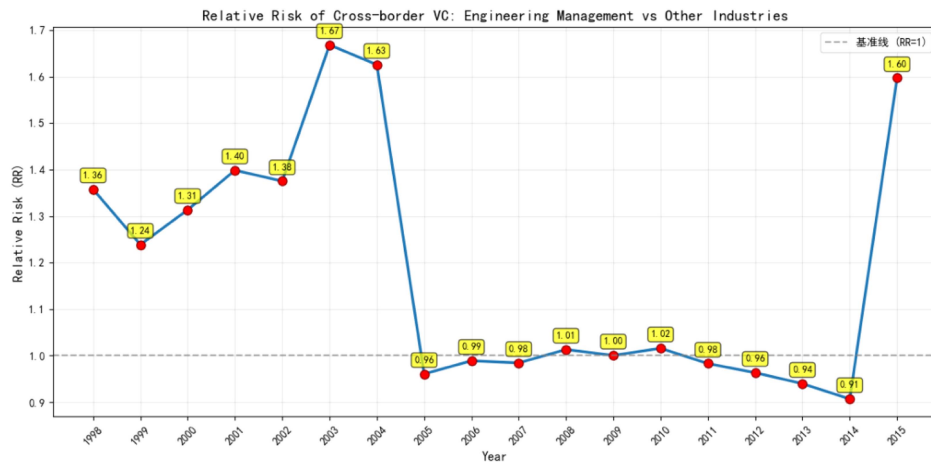


Fig. 4. Relative risk of cross-border VC: engineering management vs other industries.

frictions that may hinder firm performance. For engineering managers, the results suggest that collaboration with domestic venture capital partners can facilitate closer technological alignment and faster feedback loops. By leveraging geographically and institutionally aligned investors, engineering managers can enhance innovation efficiency, reduce project uncertainty, and improve the overall implementation of R&D initiatives.

Second, these findings also offer valuable insights for VC investors. When investors from different countries have already funded a startup, VC investors may experience lower returns. To avoid conflicts arising from differing management systems, cultures, policies, and laws, VC investors can consider investing in startups that have exclusively received funding from investors within the same nation [50]. This approach can contribute to smoother collaboration and enhance the chances of achieving desired investment outcomes.

Lastly, the findings provide support for governments in formulating relevant policies. Governments can encourage local investors to actively participate in supporting local startups. This article can be achieved by providing loan guarantees and

creating tax incentives to reduce reliance on MVC investments [51]. These measures aim to bolster the local entrepreneurial ecosystem, promote domestic innovation, and minimize the dependence on external VC funding, thus reducing the probability of the MVC's appearance.

Overall, the findings of this study offer invaluable insights for startups, investors, and governmental bodies alike, empowering them to make informed choices and implement strategic measures that cultivate an environment conducive to entrepreneurial triumph. By delving into the nuances of the entrepreneurial landscape, this research equips startups with a deeper understanding of market dynamics, consumer behavior, and emerging trends, enabling them to refine their strategies and navigate challenges with greater awareness. Moreover, governments can leverage the insights gleaned from this study to design policies and initiatives that nurture entrepreneurship and innovation. From regulatory frameworks to funding mechanisms, the findings of this research inform the development of strategies that empower entrepreneurs to thrive in an ever-evolving marketplace.

C. Managerial and Engineering Implications

The findings of this study have several implications for management and engineering practice. First, from a managerial perspective, firms backed by multicountry venture capital need to coordinate resources across borders [52]. Differences in regulatory environments, corporate governance standards, and institutional norms can create operational complexities that affect decision-making, strategic planning, and risk management. Managers should establish robust communication channels and monitoring mechanisms to align expectations with diverse VC partners.

Second, from an engineering management standpoint, multicountry VC involvement may influence the allocation and utilization of technological resources. Startups must integrate knowledge from multiple international sources while minimizing inefficiencies caused by misaligned objectives or cultural differences. Proper project management, clear technology transfer protocols, and structured R&D prioritization are essential to leverage the potential benefits of international VC investment.

Finally, the study highlights that while cross-border VC can provide access to global resources, the associated managerial and engineering challenges may offset these advantages. Practitioners and policymakers should therefore consider both the opportunities and potential costs when designing strategies for international venture capital engagement.

In Figs. 1 and 2, this diverging heatmap illustrates how the share of cross-border VC investment in each industry deviates from the overall yearly average between 1998 and 2015. The horizontal axis represents time, while the vertical axis lists industries. The color scale highlights the relative difference: red areas indicate industries that attracted a higher share of cross-border VC than the overall average in that year, while blue areas indicate industries that received relatively less.

Several patterns stand out in Fig. 3. For instance, R&D and Engineering, as well as Software & Internet, consistently show red shading during the early 2000s, suggesting these sectors were more internationally integrated in terms of VC flows. In contrast, sectors such as Chemicals and Energy & Environment are often shaded blue, indicating they lagged behind the overall trend in attracting cross-border VC. Occasional sharp deviations, such as the spike in Culture & Entertainment around 1999–2000, reveal moments of heightened international interest that did not persist.

Overall, this visualization emphasizes sectoral heterogeneity: some industries, particularly knowledge-intensive or technology-oriented ones, systematically benefited more from cross-border VC than others. This provides evidence that the globalization of venture capital is not uniform, but rather reflects sector-specific dynamics.

To examine the temporal patterns of cross-border VC allocation to engineering management sectors, we employ a relative risk (RR) metric:

$$R_t =$$

$$\frac{P(\text{Cross-border VC} \mid \text{Engineering Management, year} = t)}{P(\text{Cross-border VC} \mid \text{Other Industries, year} = t)} \quad (4)$$

where engineering management encompasses seven core sectors: Construction & real estate, R&D, and engineering.

consultancy, energy & environment, high-tech manufacturing, telecommunication services, and chemicals. These sectors represent the fundamental domains of engineering project delivery, technological innovation, and specialized technical services.

In Fig. 4, our analysis reveals distinct temporal patterns in how international venture capital targets engineering-intensive sectors. The peaked RR of 1.67 in 2003 coincides with the global infrastructure boom and increased foreign investment in engineering-driven projects. The post-2005 normalization reflects market maturation, while the 2015 resurgence aligns with emerging technologies in smart infrastructure and sustainable engineering—areas where engineering management expertise creates competitive advantages. These findings contribute to engineering management literature by demonstrating how sector-specific technical capabilities and project delivery competencies influence international investment patterns over time, particularly during periods of technological transformation and infrastructure development.

C. Limitations and Future Directions

This research aims to assess the negative influence of MVC on European startups. However, the study has some limitations that need additional investigation in future research. First, due to the restriction of data, the estimated results are based on samples mainly from Europe, so our findings may need to be revised in the United States, China, and other economies. Future research can extend the exploration of this issue by conducting estimations based on the evidence from regions other than Europe.

Moreover, our work does consider all investor-level information. Since we mainly focus on the national trait of the VCs (i.e., Multiple_{it}) and pay less attention to other investors' traits such as the investors' specific location and investors' status (e.g., active, or inactive investor). Such traits may also influence the performance of the VC-receiving firms, and remain to be investigated through more specific analyses in the future.

VI. CONCLUSION

Cross-border venture capital is often presumed to enhance entrepreneurial performance by providing global resources and market access. Our evidence suggests a more nuanced reality: in Europe from 1998 to 2015, startups backed by multicountry venture capital exhibit lower sales and ROA than those backed by single-country VC. This negative relationship is strongest in Northern and Central Europe and is most pronounced for growth- and decline-stage firms. Using a multilayer identification strategy (PSM, Heckman two-step, and 2SLS with migrant stock as an instrument) and multiple robustness checks (including CEM and variable substitutions), we confirm that the results are stable across specifications. Overall, the findings challenge the conventional “internationalization is always beneficial” view by demonstrating that, under certain institutional and developmental conditions, the coordination costs, governance frictions, and adaptation burdens embedded in MVC can outweigh its resource advantages. These results offer actionable guidance for founders selecting investors, for VCs designing syndicates, and for policymakers aiming to build resilient local entrepreneurial finance.

APPENDIX

TABLE IV

REGRESSION ESTIMATES

	Dependent variable: ln(Sales)			Dependent variable: ROA		
	(1) OLS	(2) Heckman	(3) IV	(4) OLS	(5) Heckman	(6) IV
Multiple	-0.204*** (0.037)	-0.216*** (0.038)	-1.437*** (0.491)	-0.082*** (0.009)	-0.088*** (0.009)	-0.231* (0.136)
Age	0.023*** (0.003)	0.026** (0.011)	0.044*** (0.008)	0.006*** (0.001)	0.003 (0.003)	0.008*** (0.002)
IF to TA	-1.899*** (0.078)	-1.907*** (0.080)	-1.895*** (0.082)	-0.001*** (0.000)	-0.001*** (0.000)	-0.001*** (0.000)
Shareholders' funds	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000* (0.000)	0.000 (0.000)	0.000** (0.000)
Cashflow	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
Loans	0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
IMR		-0.006 (0.346)			-0.090 (0.084)	
Constant	8.429*** (0.832)	7.833*** (0.602)	9.959*** (0.982)	0.164 (0.212)	0.041 (0.147)	-0.142 (0.240)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	20,197	19,150	19,257	22,322	21,178	21,297
R-squared	0.076	0.078	0.029	0.014	0.014	0.004

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Results from OLS, Heckman two-step, and 2SLS regressions. The key variable, Multiple_{*it*}, equals 1 if a startup received VC from multiple countries. Controls include cash flow, loans, age, and year and industry fixed effects. Columns (1)–(3) use ln(Sales); columns (4)–(6) use ROA as the dependent variable

TABLE V

OLS ESTIMATES

	Dependent variable: ln(Sales)			Dependent variable: ROA		
	(1) North	(2) Central	(3) South	(4) North	(5) Central	(6) South
Multiple	-0.251*** (0.055)	-0.169*** (0.057)	0.118 (0.129)	-0.104*** (0.014)	-0.067*** (0.013)	-0.036 (0.034)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,787	8,558	1,912	9,762	9,419	2,116

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Table V shows the results of OLS regressions. The key independent variable is the dummy variable Multiple_{*it*}, which reflects whether a startup has received VC investments from multiple nations. Meanwhile, the accounting variables such as shareholders' funds, cash flow, loans, the startup's age as well as year fixed effects and industry fixed effects are added as the control variables. Columns (1) to (3) take ln(Sales) as the proxy of the company's performance, while columns (4) to (6) take ROA, the return on assets, as the proxy variable.

TABLE VI

HECKMAN TWO-STEP ESTIMATES

	Dependent variable: ln(Sales)			Dependent variable: ROA		
	(1) North	(2) Central	(3) South	(4) North	(5) Central	(6) South
Multiple	-0.220*** (0.056)	-0.180*** (0.057)	0.087 (0.130)	-0.101*** (0.014)	-0.068*** (0.014)	-0.030 (0.034)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
IMR	0.870*** (0.244)	-0.845** (0.354)	-3.955*** (1.264)	0.090 (0.062)	-0.073 (0.082)	0.139 (0.329)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	8,728	8,508	1,853	9,695	9,365	2,043

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Table VI shows the results of Heckman two-step method. The key independent variable is the dummy variable Multiple_{*it*}, which reflects whether a startup has received VC investments from multiple nations. Meanwhile, the accounting variables such as shareholders' funds, cash flow, loans, the startup's age as well as year fixed effects and industry fixed effects are added as the control variables. Columns (1) to (3) take ln(Sales) as the proxy of the company's performance, while columns (4) to (6) take ROA, the return on assets, as the proxy variable.

TABLE VII
OLS ESTIMATES

	Dependent variable: ln(Sales)			Dependent variable: ROA		
	(1) Growth	(2) Mature	(3) Decline	(4) Growth	(5) Mature	(6) Decline
Multiple	-0.206** (0.088)	-0.100* (0.053)	-0.133** (0.066)	-0.093*** (0.020)	-0.020 (0.013)	-0.101*** (0.019)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,174	7,721	5,445	3,195	7,755	5,452

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Table VII shows the results of OLS regressions. The key independent variable is the dummy variable $Multiple_{it}$, which reflects whether a startup has received VC investments from multiple nations. Meanwhile, the accounting variables such as shareholders' funds, cash flow, loans as well as year fixed effects and industry fixed effects are added as the control variables. Columns (1) to (3) take ln(Sales) as the proxy of the company's performance, while columns (4) to (6) take ROA, the return on assets, as the proxy variable.

TABLE VIII
HECKMAN TWO-STEP ESTIMATES

	Dependent variable: ln(Sales)			Dependent variable: ROA		
	(1) Growth	(2) Mature	(3) Decline	(4) Growth	(5) Mature	(6) Decline
Multiple	-0.199** (0.091)	-0.110** (0.055)	-0.129* (0.067)	-0.101*** (0.020)	-0.022 (0.014)	-0.106*** (0.019)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
IMR	0.055 (0.429)	0.337 (0.529)	-1.350 (1.259)	0.022 (0.096)	-0.048 (0.130)	0.167 (0.364)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,038	7,315	5,187	3,056	7,348	5,194

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Table VIII shows the results of Heckman two-step method. The key independent variable is the dummy variable $Multiple_{it}$, which reflects whether a startup has received VC investments from multiple nations. Meanwhile, the accounting variables such as shareholders' funds, cash flow, loans as well as year fixed effects and industry fixed effects are added as the control variables. Columns (1) to (3) take ln(Sales) as the proxy of the company's performance, while columns (4) to (6) take ROA, the return on assets, as the proxy variable.

TABLE IX
ROBUSTNESS TEST AND FURTHER CONSIDERATION: ALTERNATIVE VARIABLE AND CONTROLLING FOR ENDOGENEITY

	Dependent variable: ln(Sales)		Dependent variable: ROA	
	(1)	(2)	(3)	(4)
Count	-0.468*** (0.102)	-0.485*** (0.105)	-0.138*** (0.025)	-0.144*** (0.026)
Control Variable	Yes	Yes	Yes	Yes
IMR		-37.769*** (2.642)		-2.928*** (0.659)
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	20,199	19,257	22,324	21,298

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The table shows the robustness checks. The key independent variable is the dummy variable $Count_{it}$, which reflects the natural logarithm of the count of unique countries from which company i received investments in year t from its historical data. Columns (1) and (3) show the results of OLS methods, and columns (2) and (4) show the results of Heckman two-step methods. Meanwhile, the accounting variables such as shareholders' funds, cash flow, the startup's age, year fixed effects, and industry fixed effects are added as the control variables.

TABLE X
ROBUSTNESS TEST AND FURTHER CONSIDERATION: COARSENEDED EXACT MATCHING (CEM)

	Dependent variable: ln(Sales)		Dependent variable: ROA	
	(1) Heckman+IV	(2) CEM+IV	(3) Heckman+IV	(4) CEM+IV
Multiple	-0.065* (0.036)	-1.117*** (0.393)	-0.043*** (0.009)	-0.221** (0.111)
Control Variables	Yes	Yes	Yes	Yes
	0.386 (0.315)		0.010 (0.076)	
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	20,572	20,679	22,225	22,346

Note: Standard errors in parentheses; Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The table shows the regression results based on CEM. Columns (1) to (2) take ln(Sales) as the proxy of the company's performance, while columns (3) to (4) take ROA, the return on assets, as the proxy variable. The accounting variables such as shareholders' funds, cash flow, loans, the startup's age as well as year fixed effects and industry fixed effects are added as the control variables. Columns (1) and (3) show the results of Heckman two-step method, while columns (2) and (4) show the results after matching using the CEM method with the inclusion of instrumental variables.

REFERENCES

- [1] A. Köhn, "The determinants of startup valuation in the venture capital context: A systematic review and avenues for future research," *Manage. Rev. Quart.*, vol. 68, no. 1, pp. 3–36, Feb. 2018, doi: [10.1007/s11301-017-0131-5](https://doi.org/10.1007/s11301-017-0131-5).
- [2] Y. Liu, Q. Zeng, B. Li, L. Ma, and J. Ordieres-Meré, "Anticipating financial distress of high-tech startups in the European Union: A machine learning approach for imbalanced samples," *J. Forecasting*, vol. 41, no. 6, pp. 1131–1155, Sep. 2022, doi: [10.1002/for.2852](https://doi.org/10.1002/for.2852).
- [3] S. Bernstein, X. Giroud, and R. R. Townsend, "The impact of venture capital monitoring," *J. Finance*, vol. 71, no. 4, pp. 1591–1622, Aug. 2016, doi: [10.1111/jofi.12370](https://doi.org/10.1111/jofi.12370).
- [4] B. Spigel and R. Harrison, "Toward a process theory of entrepreneurial ecosystems," *Strategic Entrepreneurship*, vol. 12, no. 1, pp. 151–168, Mar. 2018, doi: [10.1002/sej.1268](https://doi.org/10.1002/sej.1268).
- [5] J. Midgley and K. Tang, "Introduction: Social policy, economic growth and developmental welfare," *Int. J. Soc. Welfare*, vol. 10, no. 4, pp. 244–252, Oct. 2001, doi: [10.1111/1468-2397.00180](https://doi.org/10.1111/1468-2397.00180).
- [6] O. Colombo, "The use of signals in new-venture financing: A review and research agenda," *J. Manage.*, vol. 47, no. 1, pp. 237–259, Jan. 2021, doi: [10.1177/0149206320911090](https://doi.org/10.1177/0149206320911090).
- [7] A. Schertler and T. Tykvová, "Venture capital and internationalization," *Int. Bus. Rev.*, vol. 20, no. 4, pp. 423–439, Aug. 2011, doi: [10.1016/j.ibusrev.2010.07.009](https://doi.org/10.1016/j.ibusrev.2010.07.009).
- [8] J. Yan, S. Wang, J. Xiong, L. Scaringella, and X. Chen, "Value creation reflecting CVC strategic orientations in internet platform business ecosystems: The case of tencent," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 1531–1541, 2024, doi: [10.1109/tem.2023.3336407](https://doi.org/10.1109/tem.2023.3336407).
- [9] R. Owen, "Lessons from government venture capital funds to enable transition to a low-carbon economy: The U.K. case," *IEEE Trans. Eng. Manage.*, vol. 70, no. 3, pp. 1040–1054, Mar. 2023, doi: [10.1109/tem.2021.3094992](https://doi.org/10.1109/tem.2021.3094992).
- [10] J. J. Heckman, "Sample selection bias as a specification error," *Econometrica*, vol. 47, no. 1, pp. 153–161, Jan. 1979, doi: [10.2307/1912352](https://doi.org/10.2307/1912352).
- [11] Y. Liu and F. Wang, "Investigating the interpretability of ChatGPT in mental health counseling: An analysis of artificial intelligence generated content differentiation," *Comput. Methods Programs Biomed.*, vol. 268, Aug. 2025, Art. no. 108864, doi: [10.1016/j.cmpb.2025.108864](https://doi.org/10.1016/j.cmpb.2025.108864).
- [12] Y. Li, I. B. Vertinsky, and J. Li, "National distances, international experience, and venture capital investment performance," *J. Bus. Venturing*, vol. 29, no. 4, pp. 471–489, Jul. 2014, doi: [10.1016/j.jbusvent.2013.08.002](https://doi.org/10.1016/j.jbusvent.2013.08.002).
- [13] A. Vazirani and T. Bhattacharjee, "What motivates new venture investors? A mixed-method study approach," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 10896–10911, 2024, doi: [10.1109/TEM.2024.3404882](https://doi.org/10.1109/TEM.2024.3404882).
- [14] J. G. Covin, R. B. Bouncken, and R. P. Garrett, "Venture relatedness and learning proficiency as drivers of venture operations autonomy," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 13936–13948, 2024, doi: [10.1109/TEM.2024.3435545](https://doi.org/10.1109/TEM.2024.3435545).
- [15] D. Devigne, T. Vanacker, S. Manigart, and I. Paeleman, "The role of domestic and cross-border venture capital investors in the growth of portfolio companies," *Small Bus. Econ.*, vol. 40, no. 3, pp. 553–573, Apr. 2013, doi: [10.1007/s11187-011-9383-y](https://doi.org/10.1007/s11187-011-9383-y).
- [16] M. Narayanan and M. Levesque, "Venture capital deals: Beliefs and ownership," *IEEE Trans. Eng. Manage.*, vol. 61, no. 4, pp. 570–582, Nov. 2014, doi: [10.1109/TEM.2014.2338211](https://doi.org/10.1109/TEM.2014.2338211).
- [17] M. Guerini and A. Quas, "Governmental venture capital in Europe: Screening and certification," *J. Bus. Venturing*, vol. 31, no. 2, pp. 175–195, Mar. 2016, doi: [10.1016/j.jbusvent.2015.10.001](https://doi.org/10.1016/j.jbusvent.2015.10.001).
- [18] Y. Liu, L. Ma, Y. Dou, Z. Zhu, L. Ma, and Z. Liu, "Injecting new insights: How do review sentiment and rating inconsistency shape the helpfulness of airline reviews?," *Inf. Process. Manage.*, vol. 62, no. 4, 2025, Art. no. 104088, doi: [10.1016/j.ipm.2025.104088](https://doi.org/10.1016/j.ipm.2025.104088).
- [19] X. Dai, G. Chapman, and H. Shen, "Late-stage venture capital and firm performance: Evidence from small and medium-sized enterprises in China," *Appl. Econ.*, vol. 54, no. 20, pp. 2356–2372, Apr. 2022, doi: [10.1080/00036846.2021.1989370](https://doi.org/10.1080/00036846.2021.1989370).
- [20] S. Singh and B. S. Mungila Hillemane, "An analysis of the financial performance of tech startups: Do quantum and sources of finance make a difference in India?," *Small Enterprise Res.*, vol. 30, no. 3, pp. 374–399, Sep. 2023, doi: [10.1080/13215906.2023.2270447](https://doi.org/10.1080/13215906.2023.2270447).
- [21] D. Devigne, S. Manigart, and M. Wright, "Escalation of commitment in venture capital decision making: Differentiating between domestic and international investors," *J. Bus. Venturing*, vol. 31, no. 3, pp. 253–271, May 2016, doi: [10.1016/j.jbusvent.2016.01.001](https://doi.org/10.1016/j.jbusvent.2016.01.001).
- [22] M. Nadeem, T. Suleman, and A. Ahmed, "Women on boards, firm risk and the profitability nexus: Does gender diversity moderate the risk and return relationship?," *Int. Rev. Econ. Finance*, vol. 64, pp. 427–442, Nov. 2019, doi: [10.1016/j.iref.2019.08.007](https://doi.org/10.1016/j.iref.2019.08.007).
- [23] Y. Liu, F. Huang, L. Ma, Q. Zeng, and J. Shi, "Credit scoring prediction leveraging interpretable ensemble learning," *J. Forecasting*, vol. 43, no. 2, pp. 286–308, Mar. 2024, doi: [10.1002/for.3033](https://doi.org/10.1002/for.3033).
- [24] M. G. Colombo, M. Guerini, K. Hoisl, and N. M. Zeiner, "The dark side of signals: Patents protecting radical inventions and venture capital investments," *Res. Policy*, vol. 52, no. 5, Jun. 2023, Art. no. 104741, doi: [10.1016/j.respol.2023.104741](https://doi.org/10.1016/j.respol.2023.104741).
- [25] T. Flavin and T. O'Connor, "Reputation building and the lifecycle model of dividends," *Pac.-Basin Finance J.*, vol. 46, pp. 177–190, Dec. 2017, doi: [10.1016/j.pacfin.2017.09.006](https://doi.org/10.1016/j.pacfin.2017.09.006).
- [26] S. Koh, R. B. Durand, L. Dai, and M. Chang, "Financial distress: Lifecycle and corporate restructuring," *J. Corporate Finance*, vol. 33, pp. 19–33, Aug. 2015, doi: [10.1016/j.jcorpfin.2015.04.004](https://doi.org/10.1016/j.jcorpfin.2015.04.004).
- [27] S. Wu, L. Ning, S. Mu, and K. Chen, "The impact of governmental venture capital on firm innovation: The role of proactive R&D management," *IEEE Trans. Eng. Manage.*, vol. 72, pp. 2082–2096, 2025.
- [28] C. Wang, Y. Fang, and C. Zhang, "Mechanism and countermeasures of 'the innovator's dilemma' in business model," *J. Innov. Knowl.*, vol. 7, no. 2, Apr. 2022, Art. no. 100169, doi: [10.1016/j.jik.2022.100169](https://doi.org/10.1016/j.jik.2022.100169).
- [29] D. C. Mueller, "A life cycle theory of the firm," *J. Ind. Econ.*, vol. 20, no. 3, pp. 199–219, Jul. 1972, doi: [10.2307/2098055](https://doi.org/10.2307/2098055).
- [30] M. M. H. Chowdhury and M. Quaddus, "Supply chain resilience: Conceptualization and scale development using dynamic capability theory," *Int. J. Prod. Econ.*, vol. 188, pp. 185–204, Jun. 2017, doi: [10.1016/j.ijpe.2017.03.020](https://doi.org/10.1016/j.ijpe.2017.03.020).
- [31] A. Salvi, F. Martielli, E. Battisti, C. Forlano, and S. Kumar, "Climate change management and companies' financial performance: Exploring the moderating role of carbon credits," *Res. Int. Bus. Finance*, vol. 75, Mar. 2025, Art. no. 102785, doi: [10.1016/j.ribaf.2025.102785](https://doi.org/10.1016/j.ribaf.2025.102785).
- [32] D. H. B. Welsh, E. Kaciak, and N. Thongpapanl, "Influence of stages of economic development on women entrepreneurs' startups," *J. Bus. Res.*, vol. 69, no. 11, pp. 4933–4940, Nov. 2016, doi: [10.1016/j.jbusres.2016.04.055](https://doi.org/10.1016/j.jbusres.2016.04.055).
- [33] A. I. Goldberg, G. Cohen, and A. Fiegenbaum, "Reputation building: Small business strategies for successful venture development," *J. Small Bus. Manage.*, vol. 41, no. 2, pp. 168–186, Apr. 2003, doi: [10.1111/1540-627X.00074](https://doi.org/10.1111/1540-627X.00074).
- [34] G. Mele, G. Sansone, G. Secundo, and E. Paolucci, "Speeding up student entrepreneurship: The role of university business idea incubators," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 2364–2378, 2024, doi: [10.1109/TEM.2022.3175655](https://doi.org/10.1109/TEM.2022.3175655).
- [35] R. S. Santos and Y. Yang, "Foreign venture capital as channels of cross-border knowledge flows: Empirical evidence from U.S. critical industries," *IEEE Trans. Eng. Manage.*, vol. 72, pp. 1799–1812, 2025, doi: [10.1109/tem.2025.3565475](https://doi.org/10.1109/tem.2025.3565475).
- [36] J. Bhattacharyya and M. H. B. Subrahmanya, "Determinants of a digital start-up's access to VC financing in India: A signaling theory perspective," *Technological Forecasting Social Change*, vol. 207, Oct. 2024, Art. no. 123631, doi: [10.1016/j.techfore.2024.123631](https://doi.org/10.1016/j.techfore.2024.123631).
- [37] M. T. Wolfe and P. C. Patel, "Same difference? The impact of low-, medium-, and high-tech industries on venture performance and survival," *IEEE Trans. Eng. Manage.*, vol. 68, no. 6, pp. 1907–1918, Dec. 2021, doi: [10.1109/TEM.2019.2943703](https://doi.org/10.1109/TEM.2019.2943703).
- [38] A. Soleimani Dahaj and B. P. Cozzarin, "Government venture capital and cross-border investment," *Glob. Finance J.*, vol. 41, pp. 113–127, Aug. 2019, doi: [10.1016/j.gfj.2019.03.001](https://doi.org/10.1016/j.gfj.2019.03.001).
- [39] L. Grilli and S. Murtinu, "Government, venture capital and the growth of European high-tech entrepreneurial firms," *Res. Policy*, vol. 43, no. 9, pp. 1523–1543, Nov. 2014, doi: [10.1016/j.respol.2014.04.002](https://doi.org/10.1016/j.respol.2014.04.002).
- [40] Y. Liu, "Analyzing the effect of user-generated content on studio performance: A combined approach," *Managerial Decis. Econ.*, vol. 45, no. 4, pp. 2228–2248, Jun. 2024, doi: [10.1002/mde.4127](https://doi.org/10.1002/mde.4127).
- [41] K. Galuščák and G. Kátay, "Tax-benefit systems and differences in aggregate labour force participation: Comparative evidence from the Czech Republic and Hungary," *Econ. Syst.*, vol. 43, nos. 3/4, Sep. 2019, Art. no. 100701, doi: [10.1016/j.ecosys.2019.100701](https://doi.org/10.1016/j.ecosys.2019.100701).
- [42] J. Y. Wu, S. Opare, M. B. U. Bhuiyan, and A. Habib, "Determinants and consequences of debt maturity structure: A systematic review of the international literature," *Int. Rev. Financial Anal.*, vol. 84, Nov. 2022, Art. no. 102423, doi: [10.1016/j.irfa.2022.102423](https://doi.org/10.1016/j.irfa.2022.102423).

- [43] G. Mazeikaite, C. O'Donoghue, and D. M. Sologon, "What drives cross-country health inequality in the EU? Unpacking the role of socio-economic factors," *Social Indicators Res.*, vol. 155, no. 1, pp. 117–155, May 2021, doi: [10.1007/s11205-020-02587-2](https://doi.org/10.1007/s11205-020-02587-2).
- [44] P. Hribar and N. Yehuda, "The mispricing of cash flows and accruals at different life-cycle stages," *Contemporary Accounting Res.*, vol. 32, no. 3, pp. 1053–1072, Sep. 2015, doi: [10.1111/1911-3846.12117](https://doi.org/10.1111/1911-3846.12117).
- [45] J. Bailey and M. V. Tatikonda, "Accelerating venture milestone achievement: Examining the impact of resource acquisition timing," *IEEE Trans. Eng. Manage.*, vol. 65, no. 4, pp. 557–573, Nov. 2018, doi: [10.1109/TEM.2018.2859753](https://doi.org/10.1109/TEM.2018.2859753).
- [46] R. W. Smilor, "Managing the incubator system: Critical success factors to accelerate new company development," *IEEE Trans. Eng. Manage.*, vol. EM-34, no. 3, pp. 146–155, Aug. 1987, doi: [10.1109/TEM.1987.6498875](https://doi.org/10.1109/TEM.1987.6498875).
- [47] M. G. Colombo and L. Grilli, "On growth drivers of high-tech start-ups: Exploring the role of founders' human capital and venture capital," *J. Bus. Venturing*, vol. 25, no. 6, pp. 610–626, Nov. 2010, doi: [10.1016/j.jbusvent.2009.01.005](https://doi.org/10.1016/j.jbusvent.2009.01.005).
- [48] A. Ghezzi and A. Cavallo, "Agile business model innovation in digital entrepreneurship: Lean startup approaches," *J. Bus. Res.*, vol. 110, pp. 519–537, Mar. 2020, doi: [10.1016/j.jbusres.2018.06.013](https://doi.org/10.1016/j.jbusres.2018.06.013).
- [49] N. Fukugawa, "Division of labor between innovation intermediaries for SMEs: Productivity effects of interfirm organizations in Japan: Journal of small business management," *J. Small Bus. Manage.*, vol. 56, pp. 297–322, Mar. 2018, doi: [10.1111/jsbm.12345](https://doi.org/10.1111/jsbm.12345).
- [50] Y. Liu, X. Ding, M. Chi, J. Wu, and L. Ma, "Assessing the helpfulness of hotel reviews for information overload: A multi-view spatial feature approach," *Inf. Technol. Tourism*, vol. 26, no. 1, pp. 59–87, Mar. 2024, doi: [10.1007/s40558-023-00280-x](https://doi.org/10.1007/s40558-023-00280-x).
- [51] B. Lin and Y. Xie, "Driving green technology innovation in renewable energy: Does venture capital matter?," *IEEE Trans. Eng. Manage.*, vol. 71, pp. 6784–6795, 2024, doi: [10.1109/TEM.2023.3286978](https://doi.org/10.1109/TEM.2023.3286978).
- [52] Y. Liu, J. Huang, F. Huang, Z. Zhu, and L. Ma, "Deep learning-driven sentiment analysis for customer satisfaction optimization in airline reviews," *Ind. Manage. Data Syst.*, vol. 126, no. 3, pp. 833–864, Feb. 2026.

Yang Liu received the dual bachelor's degree in business administration and journalism from Xi'an Jiaotong University, Xi'an, China, in 2010, the MBA degree from IE Business School, Madrid, Spain, in 2015, and the joint Ph.D. degree in management science from Universidad Politécnica de Madrid, Madrid, and the University of Cambridge, Cambridge, U.K.

He is currently an Associate Professor with the School of Information Management, Wuhan University, Wuhan, China. His research interests include natural language processing, machine learning, information systems, and behavioral finance.

Yufei Qiu is currently working toward the master's student at the Zhongnan University of Economics and Law. Her primary research interests include finance.

Fangjing Bu is currently working toward the undergraduate degree in Wuhan University

Her primary research interests include finance, and applied econometrics.

Wanling Xie is currently working toward the undergraduate degree in Wuhan University, Wuhan, China

Her primary research interests include finance, and applied econometrics.

Lining Han received the bachelor's degrees in both economics and mathematics from Wuhan University, Wuhan, China, in 2010, the master's degree in economics from Shanghai Jiao Tong University, Shanghai, China, in 2013, and the Ph.D. degree in economics from the University of Hawaii at Manoa, Honolulu, HI, USA, in 2018.

He is currently an Associate Professor with Economics and Management School, Wuhan University. His research interests include microeconomics, game theory, mechanism design, and network economics.